VNR Vignana Jyothi Institute of Engineering

Computer Science and Engineering | Course Code: CS401

Artificial Intelligence & Machine Learning - Laboratory Manual & Practic

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Syllabus Focus: CS401 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Intelligent Agents & State-Space Search

" Syllabus Scope: Agent environments, BFS/DFS, heuristic A* search, IDA*, iterative deepening, admissibil

Unit 2: Adversarial Search & Constraint Satisfaction

" Syllabus Scope: Minimax game tree search, Alpha-Beta pruning, evaluation functions, Constraint Satisfac

CS401: Artificial Intelligence & Machine Learning - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Supervised Learning & Statistical Classification

" Linear regression, cost functions, gradient descent, logistic regression, decision trees ID3, C4.5, Support

Unit 4: Artificial Neural Networks & Deep Architectures

" Perceptron, multi-layer perceptron, backpropagation derivation, activation functions ReLU, Softmax, regulari

Unit 5: Modern Deep Learning & Attention Mechanisms

" Convolutional Neural Networks CNNs, Recurrent Neural Networks RNNs, LSTMs, Scaled Dot-Product Attention, Tra

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Implementation of A* Search for Solving 8-Puzzle Problem with Manhattan Distance Heuristic

Lab 2: Minimax Algorithm with Alpha-Beta Pruning for Tic-Tac-Toe Game Engine

Lab 3: Logistic Regression and Decision Tree Classifier Implementation from Scratch in Python

Lab 4: Convolutional Neural Network CNN for Image Classification using PyTorch

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Stuart Russell, Peter Norvig - Artificial Intelligence: A Modern Approach Pearson 4th Ed

* Ian Goodfellow, Yoshua Bengio - Deep Learning MIT Press

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